INCH-POUND
MIL-DTL-27072/94D
w/AMENDMENT 1
23 January 2018
SUPERSEDING
MIL-DTL-27072/94D
8 August 2006

DETAIL SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL, SINGLE CONDUCTOR, SINGLE SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27072.

Inactive for new design. For new design use MIL-DTL-55021/2.

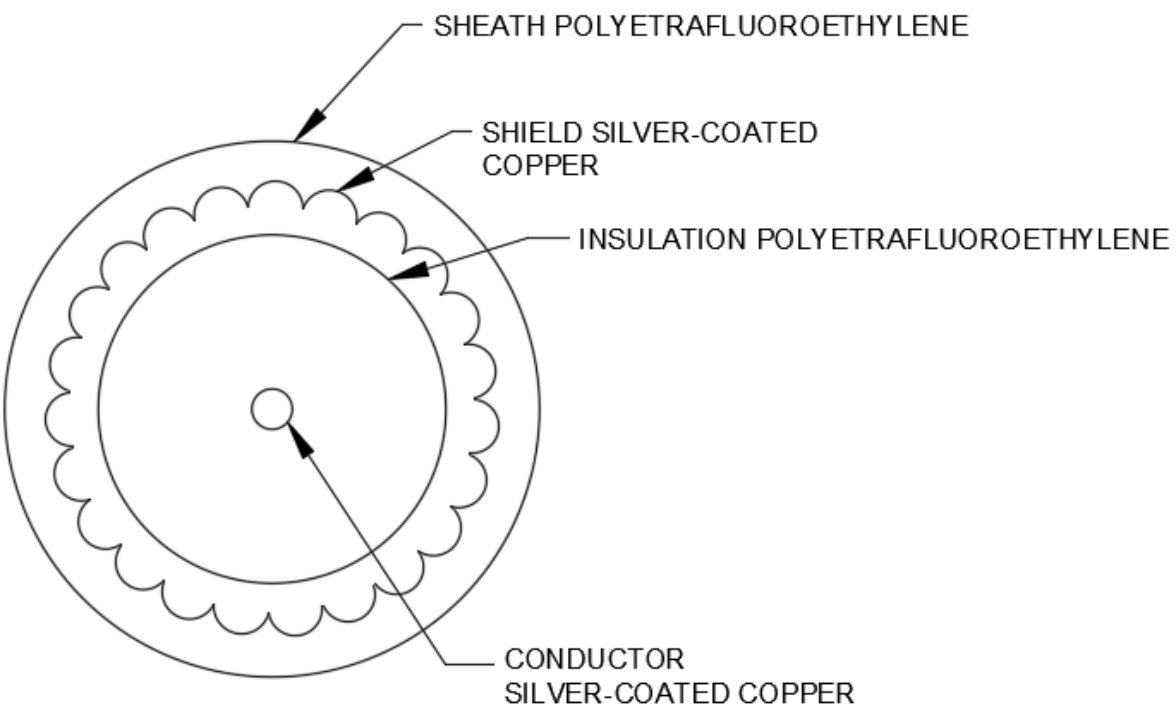



FIGURE 1. Cable configuration.

AMSC N/A

FSC 6145

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REQUIREMENTS:

| | |
|---|---|
| Design and construction: | See figure 1 |
| Number of conductors: | 1 |
| Temperature rating: | 200 °C |
| Components: | |
| Basic wire: | Either NEMA-HP3 or SAE-AS22759 (see table I) |
| Conductor: | Silver-coated copper |
| Primary insulation: | Polytetrafluoroethylene |
| Colors: | White |
| Jacket over insulation: | Not applicable |
| Filler: | Not applicable |
| Cabling: | Not applicable |
| Binder tape material: | Not applicable |
| Insulation: | Polytetrafluoroethylene extruded |
| Marking tape: | Marking tape shall be placed under the sheath or shield |
| Shield: | |
| Material: | Silver-coated copper |
| Gauge: | 38 AWG |
| Braid angle: | 20 to 40 degrees |
| Jacket under shield: | Not applicable |
| Sheath: | |
| Material: | Polytetrafluoroethylene |
| Wall thickness: | Not applicable |
| Tensile strength: | In accordance with MIL-DTL-27072 |
| Ultimate elongation: | In accordance with MIL-DTL-27072 |
| Accelerated aging: | In accordance with MIL-DTL-27072 |
| Color: | See MIL-DTL-27072 |
| Finished cable: | See table I |
| Cold bend: | Mandrel size in accordance with MIL-DTL-27072 |
| Dielectric withstanding voltage: | 2,000 V rms |
| Conductor resistance: | In accordance with basic wire specification |



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TABLE I. Construction data.

| PIN M27072/94 1/, 2/ | Wire size | Number of Strands 3/ | Strand size (AWG) | Nominal diameter Inches(mm) | Finished cable diameter inches (mm) | |
|---|---|---|---|---|---|---|
| | | | | | Minimum | Maximum |
| EDDE* | 14 | 19 | 27 | .072 (1.83) | .125 (3.18) | .152 (3.86) |
| FDDE* | 16 | 19 | 29 | .057 (1.45) | .111 (2.82) | .138 (3.51) |
| GDDE* | 18 | 19 | 30 | .050 (1.27) | .102 (2.59) | .121 (3.07) |
| JDDE* | 20 | 19** | 32 | .041 (1.04) | .095 (2.41) | .112 (2.84) |
| KDDE* | 20 | 7 | 28 | .038 (.97) | .092 (2.34) | .109 (2.77) |
| LDDE* | 22 | 19** | 34 | .032 (.81) | .086 (2.18) | .103 (2.62) |
| MDDE* | 22 | 7 | 30 | .030 (.76) | .084 (2.13) | .101 (2.57) |
| NDDE* | 24 | 19** | 36 | .035 (.89) | .076 (1.93) | .098 (2.49) |
| PDDE* | 24 | 7 | 32 | .024 (.61) | .073 (1.85) | .095 (2.41) |
| SDDE* | 26 | 7 | 34 | .019 (.48) | .068 (1.73) | .090 (2.29) |
| TDDE* | 28 | 7 | 36 | .015 (.38) | .064 (1.63) | .086 (2.18) |
| WDDE* | 30 | 7 | 38 | .012 (.30) | .061 (1.55) | .083 (2.11) |

| PIN M27072/94 1/, 2/ | Basic wire (either / or may be used) (reference) | | Formerly basic wire M16878/4 or M16878/21 (reference) |
|---|---|---|---|
| | M22759 | NEMA HP3 | |
| EDDE* | | EWBKE* | BKE9 |
| FDDE* | /11-16* | EWBJE* | BJE9 |
| GDDE* | /11-18* | EWBHE* | BHE9 |
| JDDE* | /11-20* | EWBGE* | BGE9 |
| KDDE* | - | EXBGB*<br>EWBGB* | BGB9 |
| LDDE* | /11-22* | EWBFE* | BFE9 |
| MDDE* | - | EXBFB*<br>EWBFB* | BFB9 |
| NDDE* | /11-24* | EWBEE* | BEE9 |
| PDDE* | - | EXBEB*<br>EWBEB* | BEB9 |
| SDDE* | - | EXBDB*<br>EWBDB* | BDB9 |
| TDDE* | /11-28* | EWBCB* | BCB9 |
| WDDE* | - | EXBBB*<br>EWBBB* | BBB9 |

1/ Part or Identifying Number (PIN).
2/ Asterisks (*) shall be replaced by the appropriate color designator in accordance with MIL-STD-681 and table I. If a stripe or a band is to be added, it shall be in accordance with MIL-DTL-27072.
3/ Asterisks (**) Preferred stranding number for wire size.



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Note: These constructions supersede the shielded single conductor cables of the same construction formerly covered by MIL-DTL-16878.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-27072, this document references the following:

NEMA-HP3
SAE-AS22759
MIL-STD-681
MIL-DTL-16878
MIL-DTL-55021/2

## CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2018-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, please verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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